

Tony Ko

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EDUCATION

- **University of Waterloo** Waterloo, ON
• **Candidate for Bachelor of Applied Science in Management Engineering** Sept 2025 – Apr 2030
 - **Awards:** UW CIMC & CSMC School Champion (Canadian Intermediate/Senior Mathematics Contests)
 - **Relevant Courses:** Intro to Programming (Python), Linear Algebra, Work Design and Facilities Planning, Electrical Circuits

SKILLS SUMMARY

- **Languages:** Python, SQL, JavaScript, HTML5/CSS3, MATLAB
- **Frameworks:** React Native, FastAPI, Pandas, NumPy, Matplotlib, Streamlit, Scikit-learn
- **Tools & Platforms:** Git/GitHub, RESTful APIs, PostgreSQL, SolidWorks, MS Office (Excel, PPT)
- **Soft Skills:** Leadership, Event Management, Writing, Public Speaking, Time Management

TECHNICAL PROJECTS

- **Hatched: Gamified Fitness Tracker (React Native, Python, FastAPI, PostgreSQL)**
 - Architected a full-stack mobile fitness application utilizing React Native (Expo) for the frontend and a high-performance Python (FastAPI) backend to increase user workout retention through gamification.
 - Engineered robust backend business logic and a PostgreSQL relational schema to securely process workout logs (best rep × weight), evaluate weekly target compliance, and calculate consecutive user streaks.
 - Implemented dynamic UI components and SVG animations, featuring a virtual avatar that evolves sequentially (e.g., egg to flying bird) based on sustained user engagement milestones.
- **Full-Stack Equity Dashboard (Python, Flask, JavaScript, SQLite, Chart.js, Tailwind CSS)**
 - Architected a Single-Page Application (SPA) utilizing asynchronous JavaScript (AJAX) and RESTful API endpoints to deliver real-time stock data and interactive historical price charts without page reloads.
 - Engineered a custom server-side caching layer in Python, significantly reducing reliance on the third-party **yfinance** API, eliminating rate-limiting errors, and improving data response times.
 - Designed and integrated a relational SQLite database schema to persistently store user state, allowing for dynamic CRUD operations on personalized stock watchlists.
- **Long-Term Investment Strategy & Portfolio Simulator (Python, Streamlit, NumPy)**
 - Engineered a wealth projection model to forecast **growth trajectories** over 30+ year horizons, integrating variable factors like inflation and compound return rates to simulate long-term investment outcomes.
 - Optimized a vectorized calculation engine using NumPy to process complex **scenario analyses** instantly, aiding in long-term financial stability assessment.
 - Visualized purchasing power erosion using Matplotlib, enabling users to evaluate **inflation risks** and make data-driven hedging decisions.

EXPERIENCE

- **Advanced Team Lead & Facilitator** Seongnam, South Korea
• **Culcom (Culture Community Center)** Sep 2023 – Feb 2024
 - **Leadership:** Led advanced-level group sessions, actively steering conversations to ensure curriculum alignment and maximize member participation.
 - **Member Engagement:** Fostered a supportive learning environment by leading team-building activities, which resulted in increased member participation and retention.
- **Peer Mentor** Waterloo, ON
• **Renison University College, University of Waterloo** Sep 2025 – Oct 2025
 - **Student Advisory:** Guided incoming students through university transition, providing accurate advisory on academic resources and campus policies.
 - **Cultural Integration:** Organized social activities for a diverse student body, bridging cultural gaps and enhancing the **adaptation experience** for international students.